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PhyAVBench

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PhyAVBench: A Challenging Audio Physics-Sensitivity Benchmark for Physically Grounded Text-to-Audio-Video Generation

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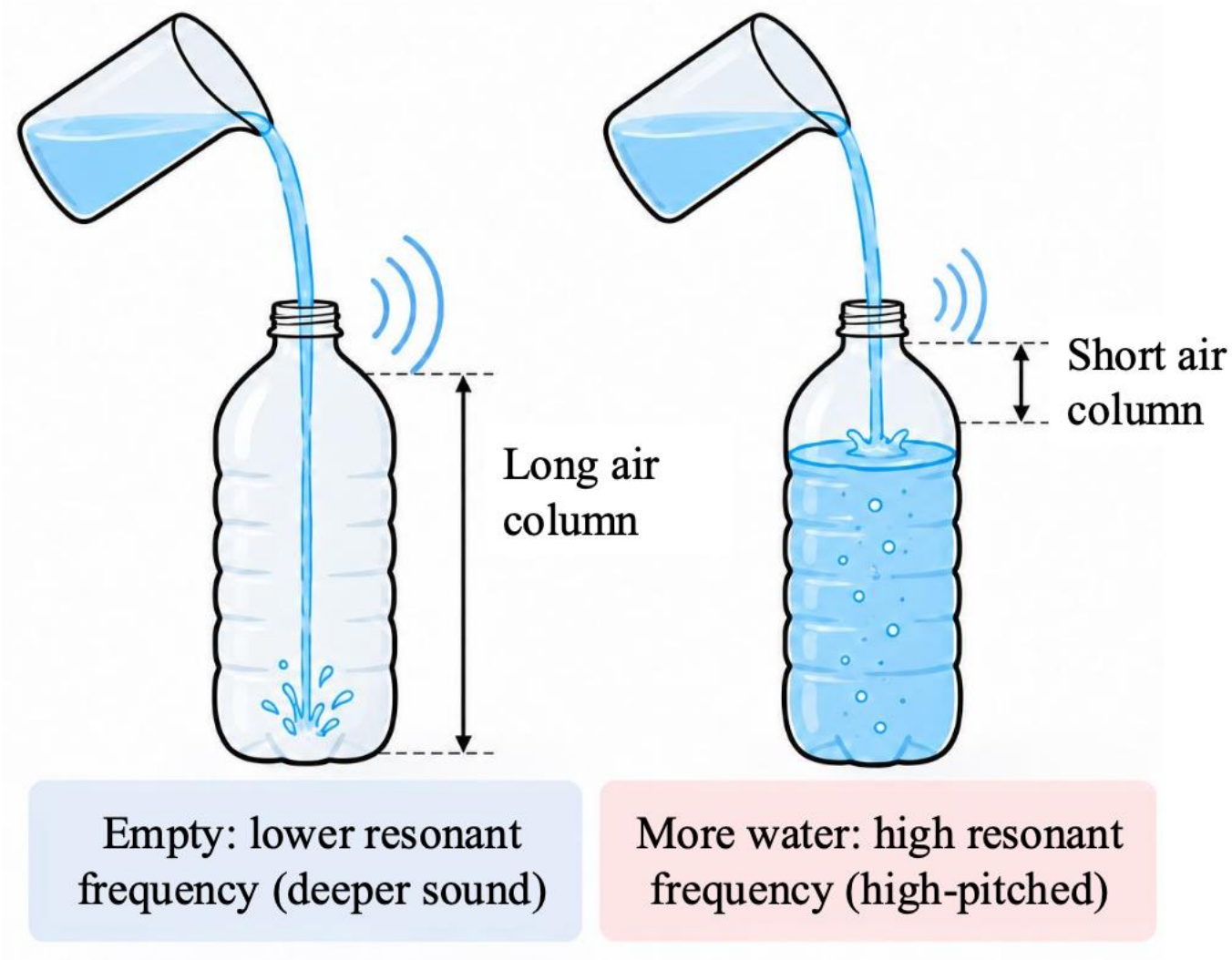
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Paper | Project | Code | Dataset

Question

Can AI-generated videos obey the physics of sound?

Example: Helmholtz Resonance



Hard example: Helmholtz resonance

Prompt A: A static medium shot shows a person holding a transparent plastic bottle. The bottle is **empty**. In a quiet indoor environment, the person brings the bottle to their mouth and blows continuously into the bottle opening for about 3 seconds, repeating this action twice.

Prompt B: A static medium shot shows a person holding a transparent plastic bottle. The bottle is **4/5 full of water**. In a quiet indoor environment, the person brings the bottle to their mouth and blows continuously into the bottle opening for about 3 seconds, repeating this action twice.



Sora2



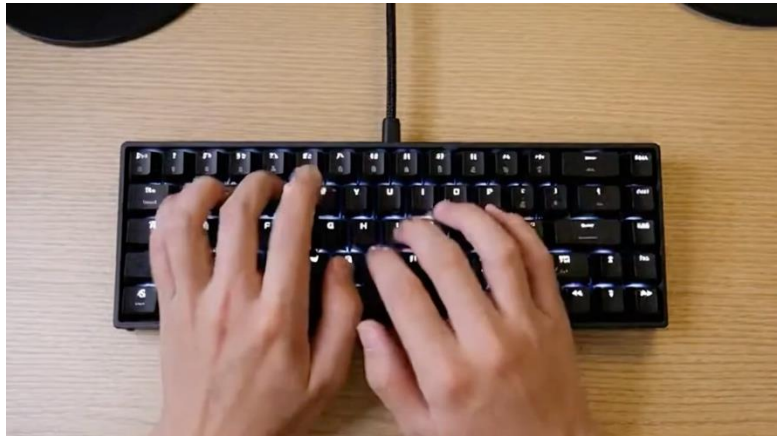


Ground Truth



Easy examples

Uncovered



Covered with plastic



Material (Sora2)

Slow



Fast



Impact Velocity (Sora2)

How to evaluate models' audio physics grounding ability?

Audio quality?

Semantic alignment?

audio-video synchronization?

The Key Idea: Test Physical Sensitivity

Controlled physical variation

Empty bottle → Water-filled bottle

Ground Truth

Reference audio change

Generated Audio

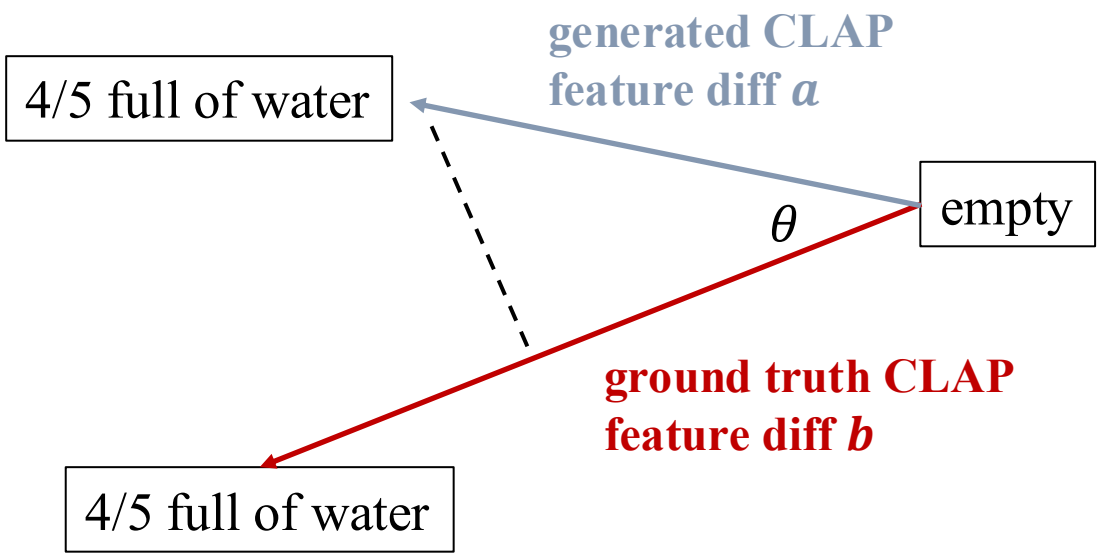
Predicted audio change

Compare direction and magnitude

Contrastive Physical Response Score (CPRS)

The Key Idea: Test Physical Sensitivity

A good model should be sensitive to physical variations.



A physically plausible pair should have

- Right feature **direction**:

$$c = \frac{\cos\theta + 1}{2} = \frac{1}{2}\left(\frac{\mathrm{a}\cdot\mathrm{b}}{\|\mathrm{a}\|\|\mathrm{b}\|} + 1\right) \in [0,1]$$

- Right feature **magnitude**:

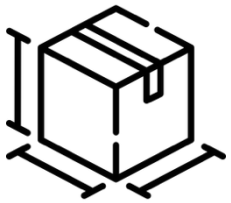
$$\mathrm{m} = \mathrm{f}\left(\frac{\mathrm{a}\cdot\mathrm{b}}{\|\mathrm{b}\|^2}\right) \in [0,1]$$

$$\mathrm{CPRS}(\mathbf{a},\mathbf{b}) = \frac{c + \mathrm{m}}{2}$$

f is a Gaussian function that maps values into range [0,1].

We built PhyAVBench for this test

- PhyAVBench offers newly recorded **PhyAV-Sound-11K**



6 Physics dimensions



41 fine-grained test points



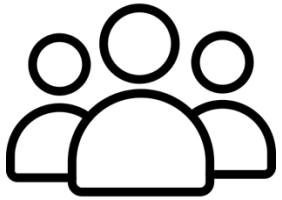
337 paired prompt groups



Reasoning annotations



11,605 videos



184 participants



25.5 hours

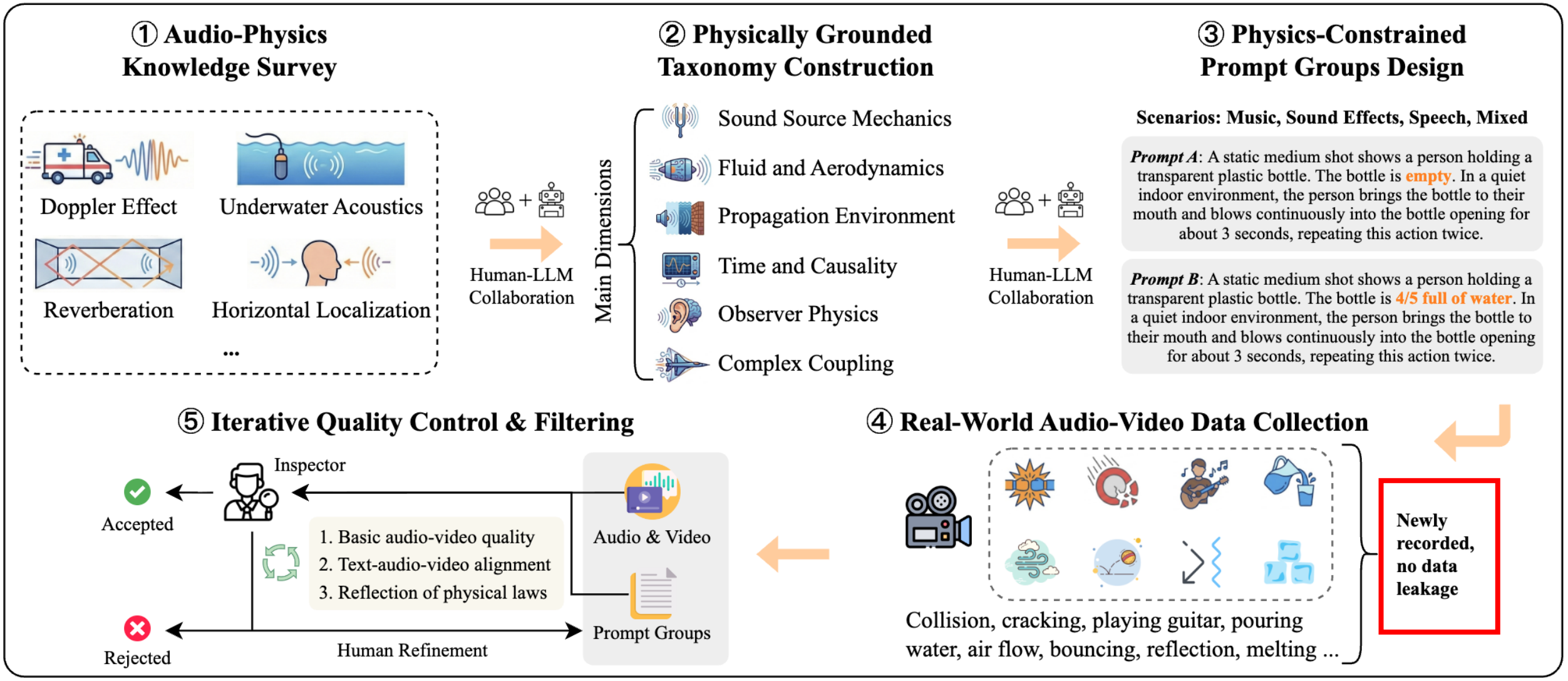


QA extension

Comparison with existing benchmarks

Evaluation	Existing benchmarks	PhyAVBench
AV synchronization	✓	✓
Controlled physical variation	✗	✓
Fine-grained physical coverage	Limited	✓
Real-world references	Limited	✓
LLM-as-a-judge	✓	✗

Data Curation Pipeline



Evaluation

Models

- 9 T2AV models
- 4 I2AV models
- 4 V2A models

Settings

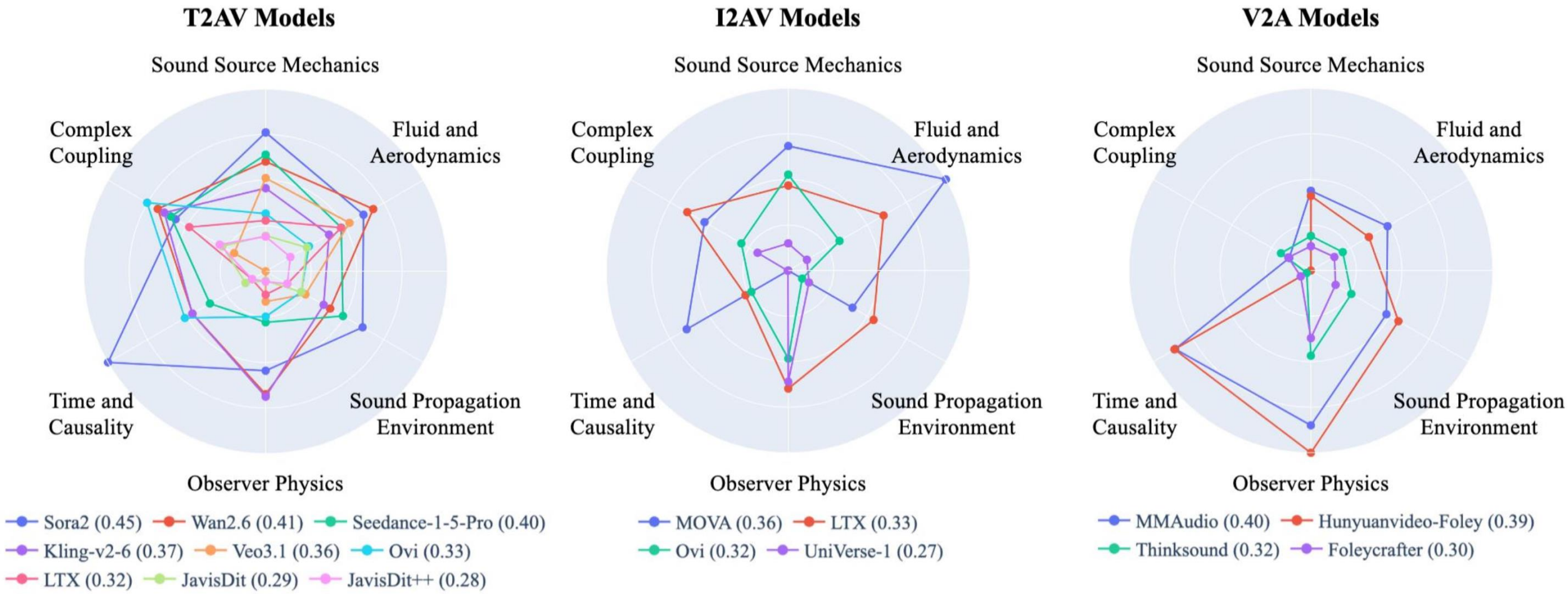
- Dataset: PhyAVSound-11K
- 337 pairs for open-source models
- 100 pairs for commercial models

Quantitative Evaluation

Table 2: Quantitative evaluation on PhyAVBench. Bold and underlined values denote the best and second-best results.

Model	AV Quality			Semantic Consistency				AV Synchronization				Physical
	FVD ↓	KVD ↓	FAD ↓	TV-IB ↑	TA-IB ↑	CLIP ↑	CLAP ↑	AV-IB ↑	DeSync ↓	AVH ↑	Javis ↑	CPRS ↑
Text-to-Audio-Video (T2AV)												
Commercial (Closed-source)												
Sora 2 [26]	**578.70**	**72.04**	4.25	0.3415	0.2028	0.2713	0.3592	0.2137	0.0370	0.2118	0.1803	**0.4512**
Wan 2.6 [43]	656.62	108.25	**1.48**	**0.3434**	0.1820	**0.2792**	0.3858	0.2143	0.2645	0.2141	0.1846	0.4120
Seedance 1.5 Pro [30]	968.03	315.38	3.09	0.3274	**0.2258**	0.2663	0.3600	**0.2794**	0.0240	0.2754	**0.2412**	0.4044
Kling v2.6 [18]	737.52	162.64	1.48	0.3373	0.2172	0.2725	**0.4252**	0.2654	0.2070	**0.2756**	0.2355	0.3718
Veo 3.1 [10]	755.51	189.83	3.48	0.3394	0.1886	0.2723	0.3393	0.2136	**0.0160**	0.2211	0.1950	0.3563
Open-source												
Ovi [23]	529.44	238.87	**2.70**	0.3113	0.0930	0.2625	0.2635	0.1509	0.3472	0.1578	0.1355	**0.3290**
LTX [11]	682.38	175.86	5.76	0.3127	0.1083	0.2619	0.2583	0.1178	**0.2531**	0.1332	0.1039	0.3235
JavisDiT [21]	656.82	244.63	3.29	0.3156	0.1350	0.2610	0.2906	0.1293	0.2893	0.1320	0.1148	0.2938
JavisDiT++ [22]	**474.62**	**76.89**	4.55	**0.3296**	**0.1506**	**0.2689**	**0.3031**	**0.1553**	0.5427	**0.1776**	**0.1498**	0.2844
Image-to-Audio-Video (I2AV)												
MOVA [34]	157.43	14.17	**1.37**	**0.3116**	**0.1859**	**0.2676**	**0.3404**	**0.2116**	0.7799	**0.2110**	**0.1679**	**0.3613**
LTX [11]	254.35	45.99	7.53	0.3100	0.1077	0.2644	0.1911	0.0964	**0.2984**	0.1010	0.0772	0.3285
Ovi [23]	**122.79**	**7.46**	3.15	0.3049	0.0911	0.2645	0.2617	0.1526	0.3410	0.1579	0.1373	0.3198
UniVerse-1 [37]	340.20	32.14	5.84	0.2573	0.0834	0.2338	0.1575	0.0792	0.7269	0.0806	0.0565	0.2729
Video-to-Audio (V2A)												
MMAudio [4]	-	-	**1.36**	0.2945	**0.2330**	0.2632	**0.3890**	0.2684	0.3165	0.2697	0.2200	**0.4003**
HunyuanVideo-Foley [31]	-	-	1.64	0.2950	0.2164	0.2636	0.3695	**0.2721**	**0.3072**	**0.2727**	**0.2260**	0.3896
ThinkSound [20]	-	-	1.60	0.2950	0.1582	0.2634	0.3412	0.1874	0.9284	0.1941	0.1458	0.3186
FoleyCrafter [50]	-	-	1.88	0.2945	0.2204	0.2627	0.3185	0.2520	0.9016	0.2580	0.2101	0.3019

Quantitative Evaluation



The audio-physics grounding performance of SOTA T2AV, I2AV, and V2A models measured by CPRS.

Human Evaluation

74 raters

20 prompt groups

Settings

Text consistency (TC)

Audio-visual synchronization (AV-Sync)

Physical variation response (PVR)

Metrics

	TC-MOS	AV-Sync-MOS	PVR-MOS
Sora 2	3.232 ±1.532	3.838 ±1.194	3.200 ±1.749
Wan 2.6	3.525 ±1.628	4.057 ±1.202	3.418 ±1.952
Seedance 1.5 Pro	2.706 ±1.481	3.473 ±1.484	2.585 ±1.765
Kling v2.6	3.014 ±1.603	3.654 ±1.361	2.914 ±1.978
Veo 3.1	2.771 ±1.483	3.408 ±1.735	2.406 ±1.696
Ovi	1.919 ±1.223	2.215 ±1.691	1.566 ±0.940
LTX	1.802 ±1.006	2.434 ±1.892	1.697 ±1.260
JavisDiT	1.430 ±0.652	1.842 ±1.391	1.456 ±0.919
JavisDiT++	1.761 ±0.967	1.883 ±1.395	1.411 ±0.798
Ground Truth	4.816 ±0.227	4.871 ±0.151	4.833 ±0.206

1.0 5.0

CPRS agrees strongly with human perception

Table 3: Pearson correlation (P.C.) across T2AV models, using per-model PVR-MOS means. For lower-is-better metrics, the score is defined as −DeSync and −FAD.

Metric	−FAD	TA-IB	CLAP	AV-IB	−DeSync	AVH	Javis	CPRS
P.C.	0.51	0.77	0.86	0.76	0.60	0.72	0.71	**0.92**

CPRS is robust across audio embedding models

Table 8: Pearson correlation (P.C.) between CPRS metrics based on three embedding models and the PVR-MOS.

Metric	LAION-CLAP	ImageBind	CAV-MAE-Sync
P.C.	0.9230	**0.9457**	0.8927

Implications for future audio-video generation

Broader coverage and scenarios
e.g., sfx, music, speech, and their mix

Dedicated judge models for audio-physics grounding

Evaluation

Physics-aware controllable generation

Incorporating explicit physical laws into generation model

Simulation-informed synthesis

Generation

Take-home Messages & Q&A

- **Beyond AV Synchronization**: Physically grounded generation requires models to respond correctly to changes in underlying physical conditions.
- **Physics-Sensitivity as a Measure**: Audio-physics sensitivity tests provide an effective way to assess physical grounding capabilities.
- **A Critical Gap Remains**: Even leading commercial systems struggle to capture fundamental audio-physical phenomena.

Paper **Project** **Code** **Dataset** **Slides**